Class 12 Mathematics

Chapter 7: Integrals

School Worksheet: step-wise solutions

20 questions **54** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

$\int e^x(\sin x + \cos x)\,dx$ is equal to:

(A) $e^x \cos x + C$

(B) $e^x \sin x + C$

(C) $-e^x \sin x + C$

(D) $e^x(\sin x - \cos x) + C$

Formula / idea: $\int e^x\,[f(x) + f'(x)]\,dx = e^x f(x) + C$

1 Compare with the standard form
$$\int e^x\,[f(x) + f'(x)]\,dx = e^x f(x) + C$$

2 Take $f(x) = \sin x$
$f'(x) = \cos x$

3 So the integral is ✓ 1
$$\int e^x(\sin x + \cos x)\,dx = e^x \sin x + C$$

Answer **(B)** $e^x \sin x + C$

Watch out: Taking $f(x) = \cos x$: then $f'(x) = -\sin x$, which does not match the bracket.

2 MCQ 1 mark

If $\int \frac{2^{1/x}}{x^2}\,dx = k \cdot 2^{1/x} + C$, then k is equal to:

(A) $-\frac{1}{\log 2}$

(B) $-\log 2$

(C) -1

(D) $\frac{1}{2}$

Formula / idea: $\int a^t\,dt = \frac{a^t}{\log a} + C$

1 Put $\frac{1}{x} = t$

$-\frac{1}{x^2}\,dx = dt$

2 Substitute

$\int \frac{2^{1/x}}{x^2}\,dx = -\int 2^t\,dt = -\frac{2^t}{\log 2} + C$

3 Put back $t = \frac{1}{x}$ and compare with $k \cdot 2^{1/x}$ ✓ 1

$-\frac{1}{\log 2} \cdot 2^{1/x} + C \;\Rightarrow\; k = -\frac{1}{\log 2}$

Answer **(A)** $-\frac{1}{\log 2}$

Watch out: Dropping the minus sign that comes from $dt = -\frac{1}{x^2}\,dx$.

3 MCQ 1 mark

The value of $\int_1^e \log x\,dx$ is:

(A) 0

(B) 1

(C) e

(D) $e \log e$

Formula / idea: $\int \log x\,dx = x\log x - x + C$ (by parts, taking $\log x \cdot 1$)

1 Integrate $\log x \cdot 1$ by parts

$\int \log x\,dx = x\log x - x$

2 Put the limits

$\left[x\log x - x\right]_1^e = (e\log e - e) - (1 \cdot \log 1 - 1)$

3 Use $\log e = 1, \log 1 = 0$
$= (e - e) - (0 - 1) = 1$ ✓ 1

Answer **(B)** 1

Watch out: Forgetting the lower limit: at $x = 1$ the value is -1, not 0.

4 MCQ 1 mark

$\int 2^{x+2}\, dx$ is equal to:

(A) $2^{x+2} + C$

(B) $2^{x+2} \log 2 + C$

(C) $\dfrac{2^{x+2}}{\log 2} + C$

(D) $2 \cdot \dfrac{2^x}{\log 2} + C$

Formula / idea: $\int a^x\, dx = \dfrac{a^x}{\log a} + C$

1 Write the formula
$\int a^x\, dx = \dfrac{a^x}{\log a} + C$

2 Here $a = 2$ and the power is $x + 2$, whose derivative is 1 ✓ 1
$\int 2^{x+2}\, dx = \dfrac{2^{x+2}}{\log 2} + C$

Answer **(C)** $\dfrac{2^{x+2}}{\log 2} + C$

Watch out: Picking (D): $2 \cdot \dfrac{2^x}{\log 2} = \dfrac{2^{x+1}}{\log 2}$, not $\dfrac{2^{x+2}}{\log 2}$.

5 MCQ 1 mark

If $\int_0^1 \frac{e^x}{1+x}\,dx = \alpha$, then $\int_0^1 \frac{e^x}{(1+x)^2}\,dx$ is equal to:

(A) $\alpha - 1 + \frac{e}{2}$

(B) $\alpha + 1 - \frac{e}{2}$

(C) $\alpha - 1 - \frac{e}{2}$

(D) $\alpha + 1 + \frac{e}{2}$

Formula / idea: $\int e^x\,[f(x) + f'(x)]\,dx = e^x f(x) + C$

1 Take $f(x) = \frac{1}{1+x}$

$f'(x) = -\frac{1}{(1+x)^2}$

2 So the difference of the two integrals is of the form $e^x[f + f']$

$\int_0^1 e^x \left[\frac{1}{1+x} - \frac{1}{(1+x)^2}\right] dx = \left[\frac{e^x}{1+x}\right]_0^1$

3 Put the limits

$\left[\frac{e^x}{1+x}\right]_0^1 = \frac{e}{2} - 1$

4 Let the required integral be J; then $\alpha - J = \frac{e}{2} - 1$ ✓ 1

$J = \alpha + 1 - \frac{e}{2}$

Answer **(B)** $\alpha + 1 - \frac{e}{2}$

Watch out: Sign slip at the last line: $J = \alpha - \left(\frac{e}{2} - 1\right)$, so the 1 becomes $+1$.

6 **Assertion-Reason** 1 mark

Assertion (A): $\int_{-1}^{1} x|x|\,dx = 0$. **Reason (R):** $f(x) = x|x|$ is an odd function, and $\int_{-a}^{a} f(x)\,dx = 0$ for every odd function f.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: If $f(-x) = -f(x)$, then $\int_{-a}^{a} f(x)\,dx = 0$

1	Check A directly: $\lvert x\rvert = -x$ on $[-1, 0]$, $\lvert x\rvert = x$ on $[0, 1]$ $\int_{-1}^{0} (-x^2)\,dx + \int_{0}^{1} x^2\,dx = -\frac{1}{3} + \frac{1}{3} = 0$	
2	So A is true.	
3	Check R: replace x by $-x$ $f(-x) = (-x)\lvert -x\rvert = -x\lvert x\rvert = -f(x)$	
4	So f is odd, and the property is the standard one. R is true.	
5	A is 0 because $x\lvert x\rvert$ is odd, so R explains A.	
6	Hence the correct option is (A).	✓ 1

Answer **(A) Both true; R is the correct explanation of A**

Watch out: Thinking $x|x|$ is even because $|x|$ is even; the factor x makes it odd.

7 **Short answer** 2 marks

Find $\int \frac{x^3 - x^2 + x - 1}{x - 1}\,dx$.

1	Factorise the numerator by grouping $x^3 - x^2 + x - 1 = x^2(x - 1) + 1(x - 1) = (x - 1)(x^2 + 1)$	✓ 1
2	Cancel $(x - 1)$, since $x \neq 1$ $\frac{x^3 - x^2 + x - 1}{x - 1} = x^2 + 1$	

3 Integrate term by term

$$\int (x^2 + 1)\,dx = \frac{x^3}{3} + x + C$$

✓ 1

Answer $\frac{x^3}{3} + x + C$

Watch out: Forgetting the constant of integration C in an indefinite integral.

Level 2

19 marks

8 **Short answer** 2 marks

Evaluate: $\int_0^{\pi/2} \sin 2x \cos 3x\,dx$.

Formula / idea: $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

1 Use $2\sin A\cos B = \sin(A+B) + \sin(A-B)$ with $A = 2x$, $B = 3x$

$$\sin 2x \cos 3x = \frac{1}{2}\left[\sin 5x + \sin(-x)\right] = \frac{1}{2}\left[\sin 5x - \sin x\right]$$

✓ ½

2 Integrate

$$I = \frac{1}{2}\left[-\frac{\cos 5x}{5} + \cos x\right]_0^{\pi/2}$$

✓ ½

3 Put the limits; $\cos\frac{5\pi}{2} = 0$, $\cos\frac{\pi}{2} = 0$, $\cos 0 = 1$

$$I = \frac{1}{2}\left[(0+0) - \left(-\frac{1}{5} + 1\right)\right]$$

✓ ½

4 Simplify

$$I = \frac{1}{2} \times \left(-\frac{4}{5}\right) = -\frac{2}{5}$$

✓ ½

Answer $-\frac{2}{5}$

Watch out: Writing $\sin(-x) = \sin x$; sine is odd, so $\sin(-x) = -\sin x$.

9 **Short answer** 2 marks

Given $\frac{d}{dx}F(x) = \frac{1}{\sqrt{2x - x^2}}$ and $F(1) = 0$, find $F(x)$.

Formula / idea: $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\frac{x}{a} + C$

1 Complete the square under the root ✓ ½
$2x - x^2 = 1 - (x^2 - 2x + 1) = 1 - (x - 1)^2$

2 Integrate using $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\frac{x}{a} + C$ ✓ ½
$F(x) = \int \frac{dx}{\sqrt{1 - (x - 1)^2}} = \sin^{-1}(x - 1) + C$

3 Use $F(1) = 0$ ✓ ½
$\sin^{-1}(0) + C = 0 \Rightarrow C = 0$

4 Hence ✓ ½
$F(x) = \sin^{-1}(x - 1)$

Answer $F(x) = \sin^{-1}(x - 1)$

Watch out: Stopping at $\sin^{-1}(x - 1) + C$ without using $F(1) = 0$ to find C.

10 **Short answer** 3 marks

Find: $\int \frac{2x - 1}{(x - 1)(x + 2)(x - 3)}\,dx$.

Formula / idea: Distinct linear factors: $\frac{px + q}{(x - a)(x - b)(x - c)} = \frac{A}{x - a} + \frac{B}{x - b} + \frac{C}{x - c}$

1 Write as partial fractions
$\frac{2x - 1}{(x - 1)(x + 2)(x - 3)} = \frac{A}{x - 1} + \frac{B}{x + 2} + \frac{C}{x - 3}$

2 Multiply both sides by the denominator
$2x - 1 = A(x + 2)(x - 3) + B(x - 1)(x - 3) + C(x - 1)(x + 2)$

3 Put $x = 1$ ✓ ½
$1 = A(3)(-2) \Rightarrow A = -\frac{1}{6}$

4 Put $x = -2$ ✓ ½
$-5 = B(-3)(-5) \Rightarrow B = -\frac{1}{3}$

5 Put $x = 3$ ✓ ½
$5 = C(2)(5) \Rightarrow C = \frac{1}{2}$

6 Integrate each fraction ✓ ½

$$I = -\frac{1}{6}\int \frac{dx}{x-1} - \frac{1}{3}\int \frac{dx}{x+2} + \frac{1}{2}\int \frac{dx}{x-3}$$

7 Use $\int \frac{dx}{x-a} = \log|x-a|$ (K = constant of integration) ✓ 1

$$I = -\frac{1}{6}\log|x-1| - \frac{1}{3}\log|x+2| + \frac{1}{2}\log|x-3| + K$$

Answer $-\frac{1}{6}\log|x-1| - \frac{1}{3}\log|x+2| + \frac{1}{2}\log|x-3| + K$

Watch out: Sign errors when putting $x = -2$: $(-2-1)(-2-3) = (-3)(-5) = 15$, positive.

11 Short answer — 3 marks

Evaluate $\int_0^{\pi/4} \log(1+\tan x)\,dx$.

Formula / idea: $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

1 Let

$$I = \int_0^{\pi/4} \log(1+\tan x)\,dx \quad \ldots(1)$$

2 Use $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ ✓ ½

$$I = \int_0^{\pi/4} \log\left[1+\tan\left(\frac{\pi}{4}-x\right)\right]dx$$

3 Expand $\tan\left(\frac{\pi}{4}-x\right)$ ✓ ½

$$1+\tan\left(\frac{\pi}{4}-x\right) = 1+\frac{1-\tan x}{1+\tan x} = \frac{2}{1+\tan x}$$

4 So ✓ ½

$$I = \int_0^{\pi/4} \log\frac{2}{1+\tan x}\,dx$$

5 Use $\log\frac{m}{n} = \log m - \log n$ and (1) ✓ ½

$$I = \int_0^{\pi/4} \log 2\,dx - \int_0^{\pi/4} \log(1+\tan x)\,dx = \frac{\pi}{4}\log 2 - I$$

6 Add I to both sides ✓ ½

$$2I = \frac{\pi}{4}\log 2$$

7 Hence

$$I = \frac{\pi}{8}\log 2$$

✓ ½

Answer $\frac{\pi}{8}\log 2$

Watch out: Writing $\tan\left(\frac{\pi}{4} - x\right) = 1 - \tan x$; use the $\tan(A - B)$ formula.

12 Short answer — 3 marks

Find $\int e^x \left(\frac{1+\sin x}{1+\cos x}\right) dx$.

Formula / idea: $\int e^x \left[f(x) + f'(x)\right] dx = e^x f(x) + C$

1 Use half-angle forms

$$1 + \cos x = 2\cos^2\frac{x}{2}, \qquad \sin x = 2\sin\frac{x}{2}\cos\frac{x}{2}$$

✓ ½

2 Split the fraction

$$\frac{1+\sin x}{1+\cos x} = \frac{1}{2\cos^2\frac{x}{2}} + \frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}} = \frac{1}{2}\sec^2\frac{x}{2} + \tan\frac{x}{2}$$

✓ 1

3 So

$$I = \int e^x \left[\tan\frac{x}{2} + \frac{1}{2}\sec^2\frac{x}{2}\right] dx$$

4 Take $f(x) = \tan\frac{x}{2}$

$$f'(x) = \frac{1}{2}\sec^2\frac{x}{2}$$

✓ ½

5 Use $\int e^x[f(x) + f'(x)]\,dx = e^x f(x) + C$

$$I = e^x \tan\frac{x}{2} + C$$

✓ 1

Answer $e^x \tan\frac{x}{2} + C$

Watch out: Missing the factor $\frac{1}{2}$ in $\frac{d}{dx}\tan\frac{x}{2} = \frac{1}{2}\sec^2\frac{x}{2}$.

13 Short answer — 3 marks

Evaluate: $\int_0^5 \left(|x-1| + |x-2| + |x-5|\right) dx$.

Formula / idea: Split the interval where the expression inside $|\ |$ changes sign.

1 Write $I = I_1 + I_2 + I_3$, one integral for each modulus term.

2 $|x-1| = 1-x$ on $[0,1]$ and $x-1$ on $[1,5]$
$$I_1 = \int_0^1 (1-x)\,dx + \int_1^5 (x-1)\,dx$$ ✓ ½

3 Integrate and put limits
$$I_1 = \left[x - \frac{x^2}{2}\right]_0^1 + \left[\frac{x^2}{2} - x\right]_1^5 = \frac{1}{2} + 8 = \frac{17}{2}$$ ✓ ½

4 $|x-2| = 2-x$ on $[0,2]$ and $x-2$ on $[2,5]$
$$I_2 = \int_0^2 (2-x)\,dx + \int_2^5 (x-2)\,dx$$ ✓ ½

5 Integrate and put limits
$$I_2 = \left[2x - \frac{x^2}{2}\right]_0^2 + \left[\frac{x^2}{2} - 2x\right]_2^5 = 2 + \frac{9}{2} = \frac{13}{2}$$ ✓ ½

6 $|x-5| = 5-x$ on all of $[0,5]$
$$I_3 = \int_0^5 (5-x)\,dx = \left[5x - \frac{x^2}{2}\right]_0^5 = \frac{25}{2}$$ ✓ ½

7 Add
$$I = \frac{17}{2} + \frac{13}{2} + \frac{25}{2} = \frac{55}{2}$$ ✓ ½

Answer $\dfrac{55}{2}$

Watch out: Splitting $|x-5|$ at $x=5$ needlessly, or using $x-5$ on $[0,5]$ where it is negative.

14 Short answer — 3 marks

Find $\int \sin^3 x \cos^2 x\,dx$.

Formula / idea: Odd power of $\sin x$: keep one $\sin x$, change the rest to $\cos x$, put $\cos x = t$.

1 Write $\sin^3 x = \sin^2 x \cdot \sin x = (1-\cos^2 x)\sin x$
$$I = \int (1-\cos^2 x)\cos^2 x\ \sin x\,dx$$ ✓ ½

2 Put $\cos x = t$
$$-\sin x\,dx = dt$$ ✓ ½

3 Substitute
$$I = -\int (1-t^2)\,t^2\,dt = -\int (t^2 - t^4)\,dt$$ ✓ ½

4 Integrate
$$I = -\frac{t^3}{3} + \frac{t^5}{5} + C$$ ✓ 1

5 Put back $t = \cos x$

$$I = -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$$

✓ ½

Answer $-\dfrac{\cos^3 x}{3} + \dfrac{\cos^5 x}{5} + C$

Watch out: Losing the minus sign from $dt = -\sin x\, dx$.

Level 3

27 marks

15 **Long answer** 5 marks

Evaluate $\displaystyle\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\, dx.$

Formula / idea: $\displaystyle\int_0^a f(x)\, dx = \int_0^a f(a - x)\, dx$ removes the factor x.

1 Let

$$I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\, dx \quad \ldots (1)$$

2 Use $\displaystyle\int_0^a f(x)\, dx = \int_0^a f(a - x)\, dx$

$$I = \int_0^{\pi} \frac{(\pi - x)\sin(\pi - x)}{1 + \cos^2(\pi - x)}\, dx$$

✓ 1

3 Use $\sin(\pi - x) = \sin x$, $\cos(\pi - x) = -\cos x$

$$I = \int_0^{\pi} \frac{(\pi - x)\sin x}{1 + \cos^2 x}\, dx \quad \ldots (2)$$

✓ ½

4 Add (1) and (2)

$$2I = \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x}\, dx$$

✓ 1

5 Put $\cos x = t$, so $-\sin x\, dx = dt$

$x = 0 \Rightarrow t = 1, \qquad x = \pi \Rightarrow t = -1$

✓ ½

6 Change the integral

$$2I = -\pi \int_1^{-1} \frac{dt}{1 + t^2} = \pi \int_{-1}^{1} \frac{dt}{1 + t^2}$$

✓ ½

7 Integrate

$$2I = \pi \left[\tan^{-1} t\right]_{-1}^{1} = \pi \left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right] = \frac{\pi^2}{2}$$

✓ 1

8 Hence

$$I = \frac{\pi^2}{4}$$

✓ ½

Answer $\frac{\pi^2}{4}$

Watch out: Forgetting to change the limits after putting $\cos x = t$.

16 **Long answer** 5 marks

Find $\int \frac{x^2+1}{(x-1)^2(x+3)}\,dx$.

Formula / idea: Repeated factor $(x-a)^2$ needs two fractions: $\frac{A}{x-a} + \frac{B}{(x-a)^2}$.

1 Write as partial fractions (repeated factor gives two terms)

$$\frac{x^2+1}{(x-1)^2(x+3)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+3}$$

✓ ½

2 Multiply both sides by the denominator

$$x^2+1 = A(x-1)(x+3) + B(x+3) + C(x-1)^2$$

✓ ½

3 Put $x = 1$

$$2 = 4B \;\Rightarrow\; B = \frac{1}{2}$$

✓ ½

4 Put $x = -3$

$$10 = 16C \;\Rightarrow\; C = \frac{5}{8}$$

✓ ½

5 Compare coefficients of x^2

$$1 = A + C \;\Rightarrow\; A = 1 - \frac{5}{8} = \frac{3}{8}$$

✓ ½

6 Integrate each fraction

$$I = \frac{3}{8}\int \frac{dx}{x-1} + \frac{1}{2}\int \frac{dx}{(x-1)^2} + \frac{5}{8}\int \frac{dx}{x+3}$$

✓ ½

7 Use $\int (x-1)^{-2}\,dx = -\frac{1}{x-1}$ (K = constant of integration)

$$I = \frac{3}{8}\log|x-1| + \frac{1}{2}\left(-\frac{1}{x-1}\right) + \frac{5}{8}\log|x+3| + K$$

✓ 1½

8 Hence

$$I = \frac{3}{8}\log|x-1| - \frac{1}{2(x-1)} + \frac{5}{8}\log|x+3| + K$$

✓ ½

Answer $\frac{3}{8}\log|x-1| - \frac{1}{2(x-1)} + \frac{5}{8}\log|x+3| + K$

Watch out: Writing $\int \frac{dx}{(x-1)^2} = \log(x-1)^2$; it is $-\frac{1}{x-1}$.

17 **Long answer** 5 marks

Evaluate $\int_0^{\pi/2} \log\sin x\, dx$.

Formula / idea: $\int_0^{2a} f(x)\,dx = 2\int_0^a f(x)\,dx$ if $f(2a-x) = f(x)$

1 Let

$$I = \int_0^{\pi/2} \log\sin x\, dx \quad \ldots(1)$$

2 Use $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

$$I = \int_0^{\pi/2} \log\sin\left(\frac{\pi}{2} - x\right) dx = \int_0^{\pi/2} \log\cos x\, dx \quad \ldots(2)$$

✓ 1

3 Add (1) and (2)

$$2I = \int_0^{\pi/2} (\log\sin x + \log\cos x)\, dx = \int_0^{\pi/2} \log(\sin x\cos x)\, dx$$

✓ ½

4 Use $\sin x\cos x = \frac{\sin 2x}{2}$

$$2I = \int_0^{\pi/2} \log\frac{\sin 2x}{2}\, dx$$

$$= \int_0^{\pi/2} \log\sin 2x\, dx - \frac{\pi}{2}\log 2$$

✓ ½

5 Let $I_1 = \int_0^{\pi/2} \log\sin 2x\, dx$. Put $2x = t$

$$dx = \frac{dt}{2}, \qquad x = 0 \Rightarrow t = 0, \qquad x = \frac{\pi}{2} \Rightarrow t = \pi$$

✓ ½

6 Change the integral

$$I_1 = \frac{1}{2}\int_0^{\pi} \log\sin t\, dt$$

✓ ½

7 Since $\sin(\pi - t) = \sin t$, use the $\int_0^{2a}$ property

$$I_1 = \frac{1}{2}\cdot 2\int_0^{\pi/2} \log\sin t\, dt = I$$

✓ 1

8 Substitute $I_1 = I$

$2I = I - \dfrac{\pi}{2}\log 2$ ✓ ½

9 Hence

$I = -\dfrac{\pi}{2}\log 2$ ✓ ½

Answer $-\dfrac{\pi}{2}\log 2$

Watch out: Treating $\displaystyle\int_0^{\pi/2} \log\sin 2x\,dx$ as I without the substitution $2x = t$ and the $\int_0^{2a}$ property.

18 **Short answer** 3 marks

Find $\displaystyle\int \frac{x+2}{\sqrt{x^2+2x+3}}\,dx.$

Formula / idea: Write numerator $= A\cdot\dfrac{d}{dx}(\text{quadratic}) + B$; then $\displaystyle\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left|x+\sqrt{x^2+a^2}\right|$

1 Write $x+2 = A\dfrac{d}{dx}(x^2+2x+3) + B$

$x+2 = A(2x+2) + B$

2 Compare coefficients: $2A = 1,\ 2A + B = 2$ ✓ ½

$A = \dfrac{1}{2}, \qquad B = 1$

3 Split the integral ✓ ½

$I = \dfrac{1}{2}\displaystyle\int \frac{2x+2}{\sqrt{x^2+2x+3}}\,dx + \int \frac{dx}{\sqrt{x^2+2x+3}} = \frac{1}{2}I_1 + I_2$

4 In I_1 put $x^2+2x+3 = t$, so $(2x+2)\,dx = dt$ ✓ ½

$I_1 = \displaystyle\int \frac{dt}{\sqrt{t}} = 2\sqrt{t} = 2\sqrt{x^2+2x+3}$

5 In I_2 complete the square ✓ ½

$x^2+2x+3 = (x+1)^2 + \left(\sqrt{2}\right)^2$

6 Use $\displaystyle\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left|x+\sqrt{x^2+a^2}\right|$ ✓ ½

$I_2 = \log\left|x+1+\sqrt{x^2+2x+3}\right|$

7 Hence ✓ ½

$I = \sqrt{x^2+2x+3} + \log\left|x+1+\sqrt{x^2+2x+3}\right| + C$

Answer $\sqrt{x^2+2x+3} + \log\left|x+1+\sqrt{x^2+2x+3}\right| + C$

Watch out: Forgetting the $\dfrac{1}{2}$ in front of I_1, which gives $2\sqrt{x^2+2x+3}$ instead of $\sqrt{x^2+2x+3}$.

19 **Long answer** 5 marks

Find $\int \left(\sqrt{\tan x} + \sqrt{\cot x}\right) dx.$

Formula / idea: Combine into $\dfrac{\sin x + \cos x}{\sqrt{\sin x \cos x}}$ and put $\sin x - \cos x = t$.

1	Write in terms of $\sin x$ and $\cos x$, then take LCM $\sqrt{\tan x} + \sqrt{\cot x} = \dfrac{\sqrt{\sin x}}{\sqrt{\cos x}} + \dfrac{\sqrt{\cos x}}{\sqrt{\sin x}} = \dfrac{\sin x + \cos x}{\sqrt{\sin x \cos x}}$	✓ 1
2	So $I = \int \dfrac{\sin x + \cos x}{\sqrt{\sin x \cos x}}\, dx$	
3	Put $\sin x - \cos x = t$ $(\cos x + \sin x)\, dx = dt$	✓ 1
4	Square $\sin x - \cos x = t$ $1 - 2\sin x \cos x = t^2 \;\Rightarrow\; \sin x \cos x = \dfrac{1 - t^2}{2}$	✓ 1
5	Substitute $I = \int \dfrac{dt}{\sqrt{\frac{1-t^2}{2}}} = \sqrt{2} \int \dfrac{dt}{\sqrt{1 - t^2}}$	✓ ½
6	Integrate $I = \sqrt{2}\, \sin^{-1} t + C$	✓ 1
7	Put back $t = \sin x - \cos x$ $I = \sqrt{2}\, \sin^{-1}(\sin x - \cos x) + C$	✓ ½

Answer $\sqrt{2}\, \sin^{-1}(\sin x - \cos x) + C$

Watch out: Putting $\sin x + \cos x = t$; its derivative is $\cos x - \sin x$, which does not match the numerator.

20 **Case study** 4 marks

Case Study: In class, a teacher demonstrates the property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, which makes many definite integrals easy to evaluate.

(i) Evaluate $\int_0^{\pi/2} \frac{\cos x}{\sin x + \cos x}\,dx$. 1 mark

(ii) Evaluate $\int_0^{\pi/2} \frac{\sin^n x}{\sin^n x + \cos^n x}\,dx$. 1 mark

(iii) Evaluate $\int_0^1 x(1-x)^{99}\,dx$. 2 marks

Formula / idea: $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$; with $a = \frac{\pi}{2}$, sin and cos swap.

(i) 1 mark

1 Let

$$I = \int_0^{\pi/2} \frac{\cos x}{\sin x + \cos x}\,dx \quad \ldots(1)$$

2 Replace x by $\frac{\pi}{2} - x$; sin and cos swap ✓ ½

$$I = \int_0^{\pi/2} \frac{\sin x}{\cos x + \sin x}\,dx \quad \ldots(2)$$

3 Add (1) and (2) ✓ ½

$$2I = \int_0^{\pi/2} 1\,dx = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

Answer (i) $\frac{\pi}{4}$

(ii) 1 mark

1 Let

$$I = \int_0^{\pi/2} \frac{\sin^n x}{\sin^n x + \cos^n x}\,dx \quad \ldots(1)$$

2 Replace x by $\frac{\pi}{2} - x$ ✓ ½

$$I = \int_0^{\pi/2} \frac{\cos^n x}{\cos^n x + \sin^n x}\,dx \quad \ldots(2)$$

3 Add (1) and (2) ✓ ½

$$2I = \int_0^{\pi/2} 1\,dx = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

Answer (ii) $\frac{\pi}{4}$

(iii) 2 marks

1 Replace x by $1-x$ (here $a=1$) ✓ ½

$$I=\int_0^1 (1-x)\,[1-(1-x)]^{99}\,dx=\int_0^1 (1-x)\,x^{99}\,dx$$

2 Multiply out ✓ ½

$$I=\int_0^1 \left(x^{99}-x^{100}\right)dx$$

3 Integrate and put limits ✓ ½

$$I=\left[\frac{x^{100}}{100}-\frac{x^{101}}{101}\right]_0^1=\frac{1}{100}-\frac{1}{101}$$

4 Simplify ✓ ½

$$I=\frac{101-100}{10100}=\frac{1}{10100}$$

Answer (iii) $\dfrac{1}{10100}$

Watch out: In (iii), expanding $(1-x)^{99}$ by the binomial theorem instead of using the property.